

Half-perimeter wirelength

Half-perimeter wirelength is the teaching yardstick for placement

The idea

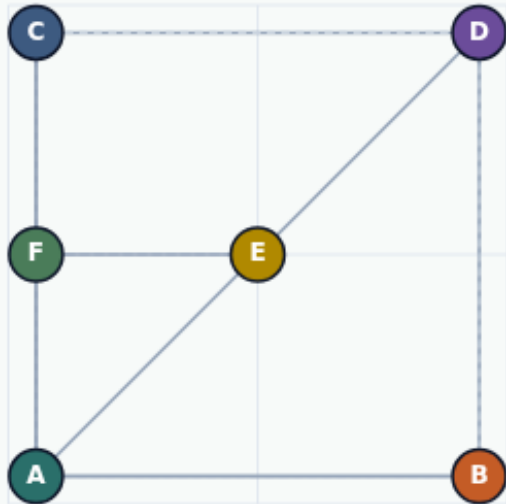
- For every net
- Sum over nets for total HPWL
- Never celebrate a tiny total that piles every cell on one point
- <!-- algorithm-walkthrough -->

Starter placement is spread out

HPWL METRICS

Starter placement is spread out

Step 1 / 5 · starter-spread · module01-01-hpwl-metrics



Explanation

Six cells A–F sit on a rough eight-by-eight canvas. The spread-out seed pulls nets long: total half-perimeter wirelength is fifty-two. This is the shared starter for every placement lab.

- Nodes = movable cells
- Dashed boxes = per-net bounding boxes
- HPWL = sum of (width + height) over nets

Total HPWL: 52
Nets: 6 (one 4-pin)
Cells: A–F

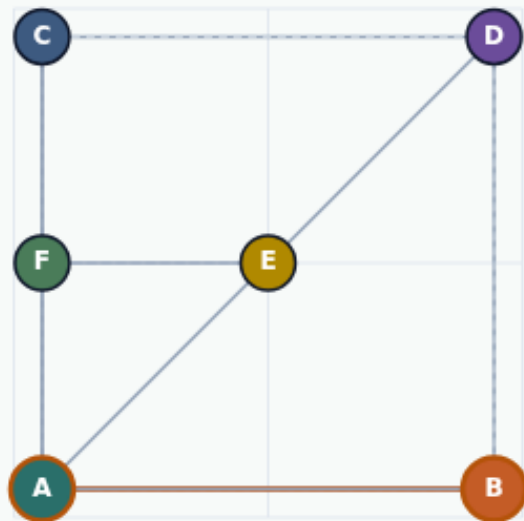
Starter placement is spread out

One net: bbox width plus height

HPWL METRICS

One net: bbox width plus height

Step 2 / 5 · one-net · module01-01-hpwl-metrics



Explanation

Net A-B alone has pins at (0,0) and (8,0). Width is eight, height is zero, so HPWL is eight. Every net uses the same bbox rule before you sum.

- min/max x and y among pins
- $HPWL(net) = (maxX - minX) + (maxY - minY)$
- Two-pin nets reduce to Manhattan span

Net A-B HPWL: 8
Net C-D HPWL: 8
4-pin ABCD HPWL: 16

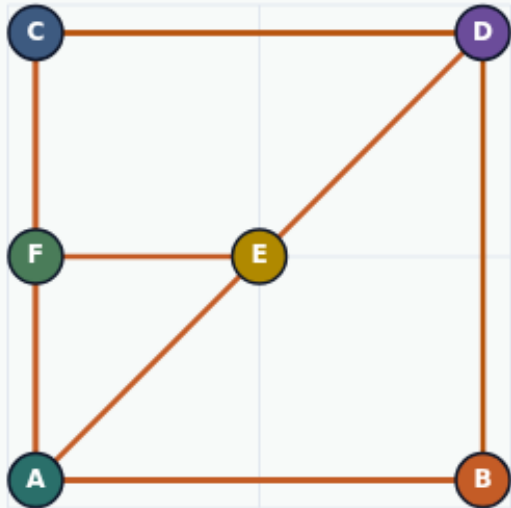
One net: bbox width plus height

Sum six nets to fifty-two

HPWL METRICS

Sum six nets to fifty-two

Step 3 / 5 · sum-nets · module01-01-hpwl-metrics



Explanation

Four two-pin nets at eight each, the four-pin ABCD net at sixteen, and E-F at four: eight times four plus sixteen plus four equals fifty-two. That is the starter golden.

- A-B, C-D, A-C, B-D: 8 each
- ABCD: 16 · E-F: 4
- Never celebrate collapse to one point

$$2\text{-pin} \times 4 = 32$$

$$4\text{-pin} = 16$$

$$E-F = 4$$

$$\text{Total} = 52$$

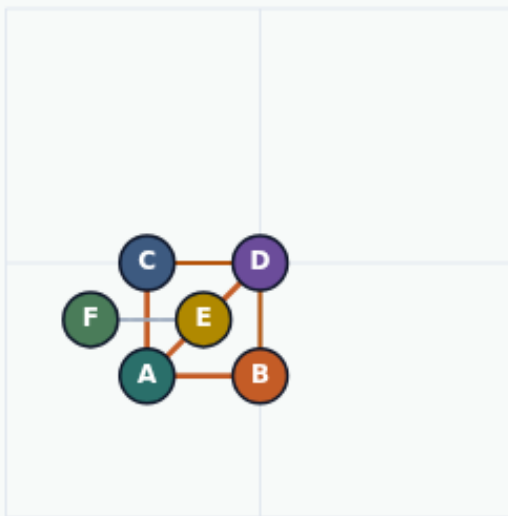
Sum six nets to fifty-two

Golden placement drops to fourteen

HPWL METRICS

Golden placement drops to fourteen

Step 4 / 5 · golden-compact · module01-01-hpwl-metrics



Explanation

The compact reference tucks A–D into a two-by-two block near the center. Same nets, shorter boxes: total HPWL falls from fifty-two to fourteen.

- Same netlist, tighter coordinates
- ABCD bbox shrinks from 16 → 4
- Reference for later algorithms

Starter HPWL: 52

Golden HPWL: 14

Delta: -38

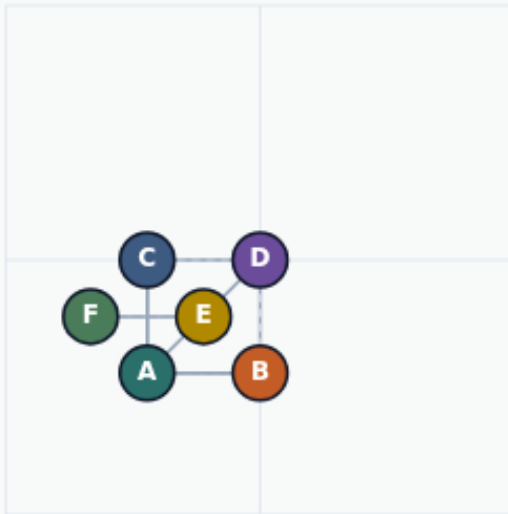
Golden placement drops to fourteen

HPWL is the teaching yardstick

HPWL METRICS

HPWL is the teaching yardstick

Step 5 / 5 · takeaway · module01-01-hpwl-metrics



Explanation

Report total HPWL before and after every move. Fifty-two versus fourteen is the starter story—but a tiny total with stacked cells is not a usable placement.

- Compute bbox HPWL per net, then sum
- Starter 52 · golden 14
- Next: clique and star models

```
starterHpwl: 52  
goldenHpwl: 14
```

HPWL is the teaching yardstick

Browser lab track

- In the browser lab track, open the **hpwl-metrics** lab from the tools shelf
- Load the starter placement, run the algorithm once
- Work the challenges that lock the goldens

Implement track

- In the implement track, open this module's examples and the course `common/`` solvers
- Parse `tiny_place.json``, run the algorithm with a deterministic seed
- Match the browser goldens before you claim the checklist

Pitfalls

- Common traps

Your turn

- Complete the checklist for at least one track, preferably both
- Implement until your metrics match the starter goldens
- When you're ready, take the short quiz, then continue to the next module